Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 8: Working with Fractions

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	8 — Working with Fractions
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This is the hardest ladder paper: every item chains several operations, so convert each mixed number to an improper fraction before you multiply or divide, flip only the divisor (never the dividend), and read 'of the remaining' as acting on the running leftover and never on the original whole; never add or subtract denominators when the denominators differ, and when you reverse a problem to 'find the original' track exactly which fraction of which quantity the known amount represents. Do not write on this paper — mark answers on your answer sheet.

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1. Compute 2 1/3 × 1 4/5 × 1 1/7 ÷ 3 3/5 and give the answer as a fraction in lowest terms.

(A) 3/4 (B) 16/9
(C) 2/3 (D) 4/3

2. A drum is 5/6 full of oil. First 2/5 of the oil in it is drawn off, then 1/4 of what then remains, then 1/3 of what is then left. What fraction of the FULL drum's capacity is now in it?

(A) 1/4 (B) 7/20
(C) 1/6 (D) 3/10

3. By what number must 4/9 be multiplied so the product is 7/9 of 2 2/5?

(A) 28/135 (B) 112/45
(C) 5/21 (D) 4 1/5

4. A man spends 3/8 of his salary on rent, then 2/5 of the remainder on food, then 1/3 of what is then left on travel, and is finally left with Rs 5000. What was his salary in rupees?

(A) 20000 (B) 18000
(C) 24000 (D) 16000

5. Which of these four fractions is the SMALLEST: 11/15, 13/18, 7/10, 4/5?

(A) 13/18 (B) 11/15
(C) 4/5 (D) 7/10

6. Compute (2 1/2 + 1 3/4) ÷ (3 1/3 – 1 5/6) and give the answer as a mixed number.

(A) 2 1/2 (B) 1 1/6
(C) 2 5/6 (D) 17/10

7. Compute 3 1/5 ÷ 1 1/3 × 2 1/4 ÷ 1 4/5 and give the answer as a mixed number.

(A) 2 1/4 (B) 3
(C) 12/5 (D) 27/8

8. On a number line, A is at 5/6 and B is at 2 1/2. A point P is placed so that AP is 3/5 of the distance AB. At what value is P?

(A) 1 1/2 (B) 1 5/6
(C) 3/2 (D) 2 1/6

9. Of a flock, 5/9 are hens. Of the hens, 3/4 are white; of the non-hens, 2/5 are white. What fraction of the WHOLE flock is white?

(A) 107/180 (B) 23/40
(C) 1/2 (D) 5/9

10. How many 2/3-litre jugs can be COMPLETELY filled from 8 3/4 litres of milk?

(A) 14 (B) 13
(C) 12 (D) 131/8

11. After giving away 2/5 of his stamps, then 1/3 of the remainder, then 1/4 of what was then left, Veer has 27 stamps. How many did he start with?

(A) 45 (B) 54
(C) 36 (D) 90

12. Arrange in ASCENDING order: 9/14, 5/8, 11/16, 2/3. Which is the correct order?

(A) 9/14 < 5/8 < 2/3 < 11/16 (B) 5/8 < 2/3 < 9/14 < 11/16
(C) 5/8 < 9/14 < 2/3 < 11/16 (D) 5/8 < 9/14 < 11/16 < 2/3

13. A vessel is 2/5 full. After 9 litres are poured in, it is 11/15 full. What is the vessel's full capacity in litres?

(A) 30 (B) 45
(C) 20 (D) 27

14. Compute 3/4 of (5 1/4 ÷ 1 1/2 × 2/3) and give the answer as a mixed number.

(A) 1 1/8 (B) 2 5/8
(C) 1 3/4 (D) 7/16

15. Compute (5/6 + 3/4) – (7/8 – 1/3) ÷ 13/24.

(A) 7/12 (B) 19/24
(C) 2/3 (D) 1

16. A garden is 8/9 of an acre. One-half of it is for tomatoes, then 3/4 of the REMAINING garden is for beans, then 2/5 of what is still left is for herbs. What fraction of an acre is for herbs?

(A) 4/45 (B) 1/9
(C) 8/135 (D) 2/45

17. By what number must 2 1/3 be multiplied so the product equals 1 5/9?

(A) 3/2 (B) 2/3
(C) 14/27 (D) 1 2/3

18. A rope is 6 1/4 m long. Pieces of 3/4 m are cut from it until no full piece remains. How many full pieces are cut and how much rope is left over?

(A) 8 pieces, 0 m left (B) 9 pieces, 1/4 m left
(C) 8 pieces, 1/4 m left (D) 7 pieces, 1/2 m left

19. Half of two-sevenths of a quantity equals three-fifths of another quantity. If the first quantity is 105, what is the second quantity?

(A) 18 (B) 50
(C) 45 (D) 25

20. Which is the LARGEST: the reciprocal of 5/8, the reciprocal of 7/4, 9/5 of 5/6, or 7/4 ÷ 5/4?

(A) the reciprocal of 5/8 (B) 7/4 ÷ 5/4
(C) 9/5 of 5/6 (D) the reciprocal of 7/4

21. Two pipes fill a tank: pipe A fills 3/10 of it per hour and pipe B fills 1/5 of it per hour. Working together, how many hours are needed to fill the whole tank?

(A) 2 (B) 10/3
(C) 5 (D) 1/2

22. Compute 5 1/4 × 2 2/9 ÷ 3 1/3 ÷ 1 3/4 and give the answer as a mixed number.

(A) 1 1/2 (B) 2
(C) 7/4 (D) 9/4

23. A jar of beads is shared so the first child takes 1/3, the second takes 2/5 of what is left, the third takes 1/2 of what then remains. If the last (fourth) child gets all that is left, namely 12 beads, how many beads were in the jar?

(A) 60
(B) 48
(C) 40
(D) 36

24. Compute (5/6 ÷ 2/3) × (3/4 ÷ 9/8) ÷ (5/9 × 3/4).

(A) 2
(B) 25/72
(C) 5/6
(D) 5/12

25. Compute 7/9 of 27/35 of 5/6 of 2 1/4 of 16.

(A) 12
(B) 9
(C) 18
(D) 24

26. Which of 13/14 and 15/14 is closer to 1, and why?

(A) 15/14, because it is larger than 1.
(B) They are equally close: each is exactly 1/14 away from 1.
(C) 13/14, because 13 is less than 15.
(D) They cannot be compared without a common denominator.

27. By what number must 4 1/5 be divided to give 7/10?

(A) 3/100
(B) 2 47/50
(C) 10/3
(D) 6

28. Compute 4 4/5 ÷ 1 1/3 ÷ 1 1/5 × 5/6 and give the answer as a mixed number.

(A) 3 3/5
(B) 1 2/3
(C) 2 1/2
(D) 18/5

29. Two-fifths of a pole is buried in soil, 1/4 of the pole is under water, and the remaining 7 m sticks out above the water. How long is the pole in metres?

(A) 35/2
(B) 20
(C) 60
(D) 28

30. Compute 9/10 ÷ (2/3 × 3/4 ÷ 5/8) and give the answer as a mixed number.

(A) 27/40
(B) 1 1/8
(C) 4/5
(D) 9/10

31. A school has 720 students. 5/8 are in the senior section. Of the seniors, 2/5 take science; of those science students, 1/2 also take computing. How many seniors take both science and computing?

(A) 180
(B) 450
(C) 90
(D) 360

32. Compute 12 ÷ 4/5 ÷ 3 ÷ 5/2 and give the answer as a mixed number.

(A) 50
(B) 2
(C) 1/2
(D) 5/4

33. A tank leaks 1/4 of its CURRENT water every hour. If it starts full, what fraction of the original water remains after 3 hours?

(A) 1/4 (B) 1/64
(C) 27/64 (D) 3/4

34. Which expression has the LARGEST value: 18 × 5/12, 18 ÷ 12/5, 18 ÷ 5/6, or 18 × 9/8?

(A) 18 × 9/8 (B) 18 × 5/12
(C) 18 ÷ 12/5 (D) 18 ÷ 5/6

35. Compute 6 2/3 ÷ 1 1/9 ÷ 4/5 × 3/8 and give the answer as a mixed number.

(A) 2 1/4 (B) 2 13/16
(C) 4 1/2 (D) 9/4

36. A juice mix is 3/10 mango pulp, 1/4 orange pulp, 1/5 pineapple pulp, and the rest water. If the mix uses 125 mL of water, how many mL is the whole mix?

(A) 400 (B) 250
(C) 625 (D) 500

37. Compute 2/3 of 1 7/8 of 4/5 of 3 1/3 and give the answer as a mixed number.

(A) 3 1/3 (B) 2 1/2
(C) 5/2 (D) 4

38. Three-fifths of a number is 12 more than one-third of the same number. What is the number?

(A) 30 (B) 60
(C) 90 (D) 45

39. A worker does 1/4 of a job on day 1. On day 2 he does 2/3 of the remaining job; on day 3 he does 3/5 of what is then left. What fraction of the whole job is still undone after day 3?

(A) 1/5 (B) 1/10
(C) 2/15 (D) 3/20

40. Which is larger and by how much: 5/6 ÷ 3/4, or 5/6 × 3/4?

(A) they are equal (B) 5/6 ÷ 3/4 is larger, by 35/72
(C) 5/6 × 3/4 is larger, by 5/8 (D) 5/6 ÷ 3/4 is larger, by 5/8

41. Compute 3 3/4 × 1 3/5 ÷ 2 1/2 × 2/3 and give the answer as a mixed number.

(A) 1 3/5 (B) 2 2/5
(C) 3 3/5 (D) 6/5

42. A cistern is filled by tap A to 5/12 of its volume and by tap B to 1/4 of its volume in one hour together. In the next hour, 2/5 of the water then present is drained. What fraction of the full cistern then remains?

(A) 1/4 (B) 11/30
(C) 2/5 (D) 1/2

43. A book has 360 pages. Maya reads 3/8 on the first day, 2/5 of the remaining on the second day, then 1/3 of what is then left on the third day. How many pages are unread after three days?

(A) 90 (B) 135
(C) 120 (D) 150

44. Half of three-fifths of a number equals one-fourth of another number. If the second number is 48, what is the first number?

(A) 20 (B) 40
(C) 80 (D) 36

45. On a number line, A is at 3/8 and B is at 1 7/8. A point P divides AB so that AP : PB = 2 : 3. At what value is P?

(A) 7/8 (B) 39/40
(C) 9/8 (D) 1 1/8

46. Compute (4 1/2 – 1 5/6) ÷ (2 1/4 – 5/6) and give the answer as a mixed number.

(A) 2 1/2 (B) 1 15/17
(C) 17/32 (D) 8/3

47. A merchant has 22 1/2 m of cloth. He sells 4/9 of it, then 3/5 of the remainder, then 1/2 of what is then left. How many metres are left?

(A) 5 (B) 7 1/2
(C) 10 (D) 2 1/2

48. Compute 8 ÷ 2/5 ÷ 4 ÷ 5/6 × 1/3 and give the answer as a mixed number.

(A) 50 (B) 12
(C) 2 (D) 3/2

49. Exactly one of the following statements about multiplying and dividing by fractions is FALSE. Which one?

(A) Dividing a number by a proper fraction gives a result larger than the number.

(B) Multiplying a number by a proper fraction gives a smaller result.

(C) Dividing a number by a fraction greater than 1 gives a result larger than the number.

(D) The reciprocal of a fraction greater than 1 is less than 1.

50. A vessel is 2/9 full. After 20 litres are added it becomes 7/9 full. What is the full capacity in litres?

(A) 36 (B) 45
(C) 30 (D) 63

51. Of a field, maize takes 2/5, then sugarcane takes 1/2 of the REMAINING field, then vegetables take 3/4 of what is still left. What fraction of the WHOLE field is left UNPLANTED after all three?

(A) 1/16 (B) 1/4
(C) 3/40 (D) 3/20

52. Compute 5 5/9 ÷ 1 2/3 ÷ 4/3 × 6/5 and give the answer as a mixed number.

(A) 2 1/2 (B) 10/3
(C) 3 (D) 5/2

53. By what number must 3 1/3 be divided so that the result is 5/6 of 2 2/5?

(A) 5/3 (B) 3/5
(C) 2 (D) 25/9

54. A photograph 6/7 m wide is enlarged to 2 1/3 times its width. By how much, in metres, does the new width exceed 1 1/2 m?

(A) 1/2 (B) 2
(C) 3/7 (D) 13/14

55. In a crate, 3/8 of the fruits are apples, 1/3 are oranges, 1/6 are pears, and the remaining 30 are mangoes. How many fruits are in the crate?

(A) 180 (B) 144
(C) 360 (D) 240

56. Half of one-third of a number, increased by one-fourth of the same number, equals 25. What is the number?

(A) 60 (B) 48
(C) 30 (D) 250

57. Compute (7/10 + 2/5) ÷ (1 1/4 – 3/5) × 5/6 and give the answer as a mixed number.

(A) 1 9/13 (B) 11/10
(C) 22/13 (D) 1 16/39

58. A field is split into four parts: the first is 2/9, the second is 1/6, the third is 1/4, and the fourth is the rest. The fourth part is what fraction of the field?

(A) 1/3 (B) 23/36
(C) 5/36 (D) 13/36

59. A vendor sells 5/12 of his oranges in the morning, then 4/7 of the remainder at noon, then 1/2 of what is then left in the evening; he is left with 30 oranges. How many oranges did he start with?

(A) 168 (B) 120
(C) 240 (D) 210

60. Order from LARGEST to SMALLEST the reciprocals of 3/7, 5/8, 9/4, and 2/3. Which is the largest of these reciprocals?

(A) the reciprocal of 2/3 (B) the reciprocal of 5/8
(C) the reciprocal of 3/7 (D) the reciprocal of 9/4

Solutions — Mathematics (Class 7), Chapter 8: Working with Fractions — Paper 5

Paper 5 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	D	21	A	31	C	41	A	51	C
2	A	12	C	22	B	32	B	42	C	52	C
3	D	13	D	23	A	33	C	43	A	53	A
4	A	14	C	24	A	34	D	44	B	54	A
5	D	15	A	25	C	35	B	45	B	55	D
6	C	16	D	26	B	36	D	46	B	56	A
7	B	17	B	27	D	37	A	47	D	57	D
8	B	18	C	28	C	38	D	48	C	58	D
9	A	19	D	29	B	39	B	49	C	59	C
10	B	20	A	30	B	40	B	50	A	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Convert all: 7/3 × 9/5 × 8/7 ÷ 18/5. Multiply the first three: (7×9×8)/(3×5×7) = 504/105 = 24/5. Then divide by 18/5 means multiply by 5/18: 24/5 × 5/18 = 24/18 = 4/3. A common slip is to flip the whole product or to forget to convert 3 3/5, which inverts the answer to its reciprocal 3/4.

2. (A) Start at 5/6. After removing 2/5, remaining = 5/6 × 3/5 = 1/2. After removing 1/4, remaining = 1/2 × 3/4 = 3/8. After removing 1/3, remaining = 3/8 × 2/3 = 6/24 = 1/4. Each 'of what remains' multiplies the running leftover, not the original 5/6; subtracting the named fractions of the whole instead would mis-state the leftover.

3. (D) The target is 7/9 × 12/5 = 84/45 = 28/15. Need 4/9 × x = 28/15, so x = (28/15) ÷ (4/9) = 28/15 × 9/4 = 252/60 = 21/5 = 4 1/5. Recheck: 28/15 × 9/4 = (28×9)/(15×4) = 252/60 = 21/5 = 4 1/5. So x = 4 1/5; multiplying instead of dividing, or mis-evaluating 7/9 of 2 2/5, gives the trap values.

4. (A) After rent, $5/8$ remains. After food, $3/5$ of that remains: $5/8 \times 3/5 = 3/8$. After travel, $2/3$ of that remains: $3/8 \times 2/3 = 1/4$ of salary = Rs 5000. So salary = 5000×4 = Rs 20000. Each later fraction acts on the running remainder; treating $2/5$ and $1/3$ as fractions of the whole salary mis-values the final leftover.

5. (D) Use the common denominator 90: $11/15 = 66/90$, $13/18 = 65/90$, $7/10 = 63/90$, $4/5 = 72/90$. The smallest numerator over 90 is $63/90 = 7/10$. Comparing by numerator alone (11 vs 13 vs 7 vs 4) or by denominator alone misranks them; only a common base is reliable.

6. (C) Sum: $5/2 + 7/4 = 10/4 + 7/4 = 17/4$. Difference: $10/3 - 11/6 = 20/6 - 11/6 = 9/6 = 3/2$. Divide: $17/4 \div 3/2 = 17/4 \times 2/3 = 34/12 = 17/6 = 2\ 5/6$. Subtracting the mixed numbers without a common denominator, or flipping the dividend, breaks the result.

7. (B) Convert: $16/5 \div 4/3 \times 9/4 \div 9/5$. Left to right: $16/5 \times 3/4 = 48/20 = 12/5$; $12/5 \times 9/4 = 108/20 = 27/5$; $27/5 \div 9/5 = 27/5 \times 5/9 = 27/9 = 3$. Only the divisor is flipped at each division; mishandling the order or flipping a multiplier gives the wrong value.

8. (B) Distance AB = $5/2 - 5/6 = 15/6 - 5/6 = 10/6 = 5/3$. AP = $3/5 \times 5/3 = 1$. So P = A + AP = $5/6 + 1 = 11/6 = 1\ 5/6$. Taking $3/5$ of B's value instead of of the gap, or measuring back from B, gives the wrong location.

9. (A) White hens: $5/9 \times 3/4 = 15/36 = 5/12$ of the flock. Non-hens = $4/9$; their white share = $4/9 \times 2/5 = 8/45$ of the flock. Use the common denominator 180: $5/12 = 75/180$ and $8/45 = 32/180$, so the total white fraction = $75/180 + 32/180 = 107/180$. Merely averaging $3/4$ and $2/5$ ignores the unequal group sizes and is the trap.

10. (B) $8\ 3/4 \div 2/3 = 35/4 \times 3/2 = 105/8 = 13\ 1/8$. Only complete jugs count, so the answer is 13 (the eighth-of-a-jug remainder cannot fill another full jug). Rounding $13\ 1/8$ up to 14, or keeping the improper fraction $105/8$ as the count, are the standard errors.

11. (D) Fraction left = $3/5 \times 2/3 \times 3/4 = (3\times2\times3)/(5\times3\times4) = 18/60 = 3/10$. So $3/10$ of the start = 27, giving start = $27 \times 10/3 = 90$. Subtracting $2/5$, $1/3$ and $1/4$ of the whole separately instead of compounding the running remainders gives a wrong total.

12. (C) Convert to a common denominator 336: $9/14 = 216/336$, $5/8 = 210/336$, $11/16 = 231/336$, $2/3 = 224/336$. Ascending by numerator: $210 < 216 < 224 < 231$, i.e. $5/8 < 9/14 < 2/3 < 11/16$. Ranking by raw numerators or denominators alone reorders them wrongly.

13. (D) The 9 litres raise the level from $2/5$ to $11/15$: rise = $11/15 - 2/5 = 11/15 - 6/15 = 5/15 = 1/3$ of capacity. So $1/3$ of capacity = 9, capacity = 27. Using $11/15$ alone, or adding the fractions instead of subtracting, gives the wrong base.

14. (C) Inside: $5\ 1/4 = 21/4$. $21/4 \div 3/2 = 21/4 \times 2/3 = 42/12 = 7/2$. Then $7/2 \times 2/3 = 14/6 = 7/3$. Now $3/4$ of $7/3 = 3/4 \times 7/3 = 21/12 = 7/4 = 1\ 3/4$. Doing the 'of' before the bracketed division, or flipping the wrong factor, changes the result.

15. (A) Division before subtraction: 7/8 − 1/3 = 21/24 − 8/24 = 13/24. Then 13/24 ÷ 13/24 = 1. Now 5/6 + 3/4 = 10/12 + 9/12 = 19/12. Finally 19/12 − 1 = 19/12 − 12/12 = 7/12. So the value is 7/12; treating the whole expression left-to-right (ignoring that ÷ binds before −) gives a different answer.

16. (D) Start 8/9. After tomatoes (1/2 used), remaining = 8/9 × 1/2 = 4/9. After beans (3/4 used), remaining = 4/9 × 1/4 = 1/9. Herbs use 2/5 of that: 1/9 × 2/5 = 2/45 of an acre. Each step's fraction acts on the running remainder; applying 3/4 or 2/5 to the original 8/9 misstates the herb area.

17. (B) Need 7/3 × x = 14/9, so x = (14/9) ÷ (7/3) = 14/9 × 3/7 = 42/63 = 2/3. Dividing 1 5/9 by 2 1/3 means flipping the divisor 7/3; multiplying instead gives 7/3 × 14/9 = 98/27, the trap.

18. (C) 6 1/4 ÷ 3/4 = 25/4 × 4/3 = 25/3 = 8 1/3. So 8 full pieces are cut, using 8 × 3/4 = 6 m, leaving 6 1/4 − 6 = 1/4 m. The leftover is the fraction of a piece (1/3 of a 3/4 m piece = 1/4 m), not 1/3 m; rounding the count up to 9 overcounts.

19. (D) Half of 2/7 of 105 = 1/2 × 2/7 × 105 = 1/7 × 105 = 15. This equals 3/5 of the second, so 3/5 × second = 15, second = 15 × 5/3 = 25. Forgetting to invert the 3/5 (multiplying 15 by 3/5 instead) gives 9; mishandling the 'half' gives 50.

20. (A) Reciprocal of 5/8 = 8/5 = 1.6. Reciprocal of 7/4 = 4/7 ≈ 0.57. 9/5 of 5/6 = 9/5 × 5/6 = 9/6 = 3/2 = 1.5. 7/4 ÷ 5/4 = 7/4 × 4/5 = 7/5 = 1.4. The largest is 8/5 = 1.6, the reciprocal of 5/8. The reciprocal of 7/4 is a proper-fraction value below 1 and is in fact the smallest of the four; assuming the largest fraction gives the largest reciprocal is exactly backwards.

21. (A) Together they fill 3/10 + 1/5 = 3/10 + 2/10 = 5/10 = 1/2 of the tank per hour. To fill 1 whole tank: 1 ÷ 1/2 = 2 hours. Adding the per-hour rates and then forgetting to invert (reporting 1/2 as the time) is the trap.

22. (B) Convert: 21/4 × 20/9 ÷ 10/3 ÷ 7/4. First 21/4 × 20/9 = 420/36 = 35/3. Then 35/3 ÷ 10/3 = 35/3 × 3/10 = 105/30 = 7/2. Then 7/2 ÷ 7/4 = 7/2 × 4/7 = 28/14 = 2. Flip only each divisor; one missed inversion derails the chain.

23. (A) After first: 2/3 left. After second: 2/3 × 3/5 = 2/5 left. After third: 2/5 × 1/2 = 1/5 left. The fourth child gets that 1/5 = 12 beads, so jar = 12 × 5 = 60. Each share acts on the running remainder; computing fractions of the original jar throughout misstates the final leftover.

24. (A) First bracket: 5/6 ÷ 2/3 = 5/6 × 3/2 = 15/12 = 5/4. Second: 3/4 ÷ 9/8 = 3/4 × 8/9 = 24/36 = 2/3. Third (the divisor): 5/9 × 3/4 = 15/36 = 5/12. Now 5/4 × 2/3 = 10/12 =

5/6; then 5/6 ÷ 5/12 = 5/6 × 12/5 = 60/30 = 2. Treating the last bracket as a multiplier rather than a divisor gives 5/6 × 5/12 = 25/72, the trap.

25. (C) Multiply across: 7/9 × 27/35 × 5/6 × 9/4 × 16. Step by step: 7/9 × 27/35 = 189/315 = 3/5; 3/5 × 5/6 = 15/30 = 1/2; 1/2 × 9/4 = 9/8; 9/8 × 16 = 144/8 = 18. Cancelling pairwise throughout, and multiplying the whole-number 16 only at the end, keeps the arithmetic clean; dropping any factor lowers the result.

26. (B) Both have denominator 14. The distance of 13/14 from 1 is 1 − 13/14 = 1/14; the distance of 15/14 from 1 is 15/14 − 1 = 1/14. The distances are equal, so the two fractions are equally close to 1. The reasoning that 'larger means closer' or 'smaller numerator means closer' both ignore that closeness is measured by distance, which is 1/14 on each side.

27. (D) Need (21/5) ÷ x = 7/10, so x = (21/5) ÷ (7/10) = 21/5 × 10/7 = 210/35 = 6. Multiplying 4 1/5 by 7/10 instead of dividing gives 21/5 × 7/10 = 147/50 = 2 47/50, the trap.

28. (C) Convert: 24/5 ÷ 4/3 ÷ 6/5 × 5/6. 24/5 ÷ 4/3 = 24/5 × 3/4 = 72/20 = 18/5; 18/5 ÷ 6/5 = 18/5 × 5/6 = 90/30 = 3; then 3 × 5/6 = 15/6 = 5/2 = 2 1/2. Carelessly treating the final × 5/6 as a division gives 3 ÷ 5/6 = 18/5 = 3 3/5, the trap value.

29. (B) Buried + under water = 2/5 + 1/4 = 8/20 + 5/20 = 13/20 of the pole. The part sticking out = 1 − 13/20 = 7/20 of the pole = 7 m. So pole = 7 ÷ (7/20) = 7 × 20/7 = 20 m. Adding 2/5 and 1/4 as 3/9 (adding denominators) is the classic error.

30. (B) Inside, left to right: 2/3 × 3/4 = 6/12 = 1/2; then 1/2 ÷ 5/8 = 1/2 × 8/5 = 8/10 = 4/5. Now 9/10 ÷ 4/5 = 9/10 × 5/4 = 45/40 = 9/8 = 1 1/8. Doing the inner operations out of order, or flipping the outer dividend, changes the result.

31. (C) Seniors = 5/8 × 720 = 450. Science seniors = 2/5 × 450 = 180. Both science and computing = 1/2 × 180 = 90. Equivalently the chained product 5/8 × 2/5 × 1/2 × 720 = 1/8 × 720 = 90. Each fraction must act on the running subgroup, not on the whole school of 720; doing so overcounts.

32. (B) Left to right: 12 ÷ 4/5 = 12 × 5/4 = 15; 15 ÷ 3 = 5; 5 ÷ 5/2 = 5 × 2/5 = 2. Only divisors are flipped; treating 3 as 1/3 or forgetting an inversion changes the value. The answer is the whole number 2.

33. (C) Each hour, 3/4 of the current water remains. After 3 hours: (3/4)^3 = 27/64. Subtracting 1/4 three times (giving 1/4 of the original) is wrong because each hour's loss is of the smaller current amount, not of the original.

34. (D) Compute each: 18 × 5/12 = 90/12 = 7.5; 18 ÷ 12/5 = 18 × 5/12 = 7.5; 18 ÷ 5/6 = 18 × 6/5 = 108/5 = 21.6; 18 × 9/8 = 162/8 = 20.25. The largest is 18 ÷ 5/6 = 21.6, because dividing by a proper fraction enlarges most here. Assuming the multiplication by 9/8 wins, or that 18 × 5/12 equals 18 ÷ 12/5 being largest, misses that division by 5/6 grows fastest.

35. (B) Convert: 20/3 ÷ 10/9 ÷ 4/5 × 3/8. 20/3 × 9/10 = 180/30 = 6; 6 ÷ 4/5 = 6 × 5/4 = 30/4 = 15/2; 15/2 × 3/8 = 45/16 = 2 13/16. A single missed flip on the 4/5 division collapses the chain to a wrong value.

36. (D) Non-water fraction = 3/10 + 1/4 + 1/5. LCD 20: 6/20 + 5/20 + 4/20 = 15/20 = 3/4. Water = 1 − 3/4 = 1/4 of the mix = 125 mL. So whole mix = 125 × 4 = 500 mL. Adding the three numerators over a wrong common denominator, or forgetting to subtract from 1, gives the wrong base.

37. (A) Convert: 2/3 × 15/8 × 4/5 × 10/3. Step: 2/3 × 15/8 = 30/24 = 5/4; 5/4 × 4/5 = 20/20 = 1; 1 × 10/3 = 10/3 = 3 1/3. Cancelling pairwise keeps the numbers small; multiplying naively then reducing risks slips.

38. (D) Let the number be n. 3/5 n − 1/3 n = 12. Common denominator 15: 9/15 n − 5/15 n = 4/15 n = 12, so n = 12 × 15/4 = 45. Subtracting the fractions wrongly (e.g. 3/5 − 1/3 = 2/2) or solving 3/5 n = 12 ignores the comparison structure.

39. (B) After day 1, remaining = 3/4. After day 2, remaining = 3/4 × 1/3 = 1/4. After day 3, remaining = 1/4 × 2/5 = 2/20 = 1/10. Each day's fraction is of the running remainder; subtracting the named fractions of the whole job mis-states what is left.

40. (B) 5/6 ÷ 3/4 = 5/6 × 4/3 = 20/18 = 10/9. 5/6 × 3/4 = 15/24 = 5/8. Difference = 10/9 − 5/8. LCD 72: 80/72 − 45/72 = 35/72. So division gives the larger value, exceeding the product by 35/72. Dividing by a proper fraction enlarges while multiplying shrinks; assuming they are equal ignores this.

41. (A) Convert: 15/4 × 8/5 ÷ 5/2 × 2/3. 15/4 × 8/5 = 120/20 = 6; 6 ÷ 5/2 = 6 × 2/5 = 12/5; 12/5 × 2/3 = 24/15 = 8/5 = 1 3/5. Flip only the divisor 5/2; treating the final 2/3 as a divisor changes the answer.

42. (C) After hour 1: 5/12 + 1/4 = 5/12 + 3/12 = 8/12 = 2/3 full. Draining 2/5 of that present water leaves 3/5 of it: 2/3 × 3/5 = 6/15 = 2/5. Draining 2/5 of the full cistern instead of 2/5 of the water present overstates the loss.

43. (A) Day 1 read 3/8 × 360 = 135; remaining 225. Day 2 read 2/5 × 225 = 90; remaining 135. Day 3 read 1/3 × 135 = 45; remaining 135 − 45 = 90. Alternatively, fraction unread = 5/8 × 3/5 × 2/3 = 30/120 = 1/4; 1/4 × 360 = 90. Applying 2/5 or 1/3 to the original 360 instead of the running remainder mis-counts.

44. (B) One-fourth of 48 = 12. So half of 3/5 of the first = 12, i.e. 3/10 of the first = 12, first = 12 × 10/3 = 40. Solving 3/5 of first = 12 (forgetting the 'half') gives 20; multiplying by 3/10 instead of dividing gives 18/5.

45. (B) AB = 15/8 − 3/8 = 12/8 = 3/2. AP : PB = 2 : 3 means AP = 2/5 of AB = 2/5 × 3/2 = 3/5. So P = A + AP = 3/8 + 3/5. LCD 40: 15/40 + 24/40 = 39/40. Taking AP as 2/3 of AB, or measuring PB instead of AP, mislocates P.

46. (B) Numerator: 9/2 − 11/6 = 27/6 − 11/6 = 16/6 = 8/3. Denominator: 9/4 − 5/6 = 27/12 − 10/12 = 17/12. Divide: 8/3 ÷ 17/12 = 8/3 × 12/17 = 96/51 = 32/17 = 1 15/17. Flipping the wrong term, or subtracting the mixed numbers without common denominators, breaks the result.

47. (D) After selling 4/9, remaining = 5/9 × 45/2 = 25/2 = 12.5 m. After 3/5 of that, remaining = 2/5 × 25/2 = 5 m. After 1/2 of that, remaining = 1/2 × 5 = 5/2 = 2 1/2 m. Each later fraction is of the running remainder; subtracting 4/9, 3/5, 1/2 of the original 22 1/2 m separately mis-counts.

48. (C) Left to right: 8 ÷ 2/5 = 8 × 5/2 = 20; 20 ÷ 4 = 5; 5 ÷ 5/6 = 5 × 6/5 = 6; 6 × 1/3 = 2. Flip each divisor; the multiplier 1/3 is not flipped. A misplaced inversion or treating ÷4 as ÷(1/4) ruins the chain. The answer is the whole number 2.

49. (C) Dividing by a fraction greater than 1 (e.g. ÷ 3/2 = × 2/3) shrinks the number, so that statement is false. The other three are true: dividing by a proper fraction enlarges, multiplying by a proper fraction shrinks, and the reciprocal of a fraction above 1 lies below 1.

50. (A) The 20 litres raise the level from 2/9 to 7/9: rise = 7/9 − 2/9 = 5/9 of capacity = 20. So capacity = 20 × 9/5 = 36. Using 7/9 alone, or 2/9 alone, instead of the difference gives the wrong base.

51. (C) Start with the whole field (1). After maize (2/5 used), remaining = 3/5. After sugarcane (1/2 used), remaining = 3/5 × 1/2 = 3/10. After vegetables (3/4 used), remaining = 3/10 × 1/4 = 3/40 of the whole field. Each fraction acts on the running remainder; applying 1/2 or 3/4 to the original whole, or reporting only the last factor 1/4, gives the wrong leftover.

52. (C) Convert: 50/9 ÷ 5/3 ÷ 4/3 × 6/5. 50/9 × 3/5 = 150/45 = 10/3; 10/3 × 3/4 = 30/12 = 5/2; 5/2 × 6/5 = 30/10 = 3. Each division flips its divisor; the final 6/5 is a multiplier. A missed inversion or treating × 6/5 as ÷ derails it.

53. (A) The target = 5/6 × 12/5 = 60/30 = 2. So 10/3 ÷ x = 2, giving x = 10/3 ÷ 2 = 10/3 × 1/2 = 10/6 = 5/3. Computing 5/6 of 2 2/5 wrongly, or multiplying 3 1/3 by 2 instead of dividing, gives the wrong number.

54. (A) New width = 6/7 × 7/3 = 42/21 = 2 m. Excess over 1 1/2 m = 2 − 3/2 = 1/2 m. Multiplying 6/7 by 2 1/3 wrongly, or comparing against the original width instead of 1 1/2 m, gives the wrong excess.

55. (D) Named fraction = 3/8 + 1/3 + 1/6. LCD 24: 9/24 + 8/24 + 4/24 = 21/24 = 7/8. Mangoes = 1 − 7/8 = 1/8 of the crate = 30. So crate = 30 × 8 = 240. Adding the three numerators over a wrong common denominator, or forgetting to subtract from 1, gives the wrong total.

56. (A) Half of 1/3 of n = 1/6 n. Adding 1/4 n: 1/6 n + 1/4 n. LCD 12: 2/12 n + 3/12 n = 5/12 n = 25, so n = 25 × 12/5 = 60. Adding the fractions wrongly as 1/6 + 1/4 = 2/10 = 1/5 (adding denominators) gives 125; forgetting the 'half' mishandles the first term.

57. (D) Sum: 7/10 + 4/10 = 11/10. Difference: 5/4 − 3/5 = 25/20 − 12/20 = 13/20. Divide: 11/10 ÷ 13/20 = 11/10 × 20/13 = 220/130 = 22/13. Then × 5/6: 22/13 × 5/6 = 110/78 = 55/39 = 1 16/39. The bracket sums come first, only the divisor 13/20 is flipped, and the final × 5/6 is a multiplier.

58. (D) First three sum to 2/9 + 1/6 + 1/4. LCD 36: 8/36 + 6/36 + 9/36 = 23/36. Fourth = 1 − 23/36 = 13/36. Adding numerators over a wrong denominator, or forgetting to subtract from the whole, gives the wrong leftover.

59. (C) Fraction left = 7/12 × 3/7 × 1/2 = (7×3×1)/(12×7×2) = 21/168 = 1/8. So 1/8 of the start = 30, start = 30 × 8 = 240. Each later fraction acts on the running remainder; subtracting 5/12, 4/7, 1/2 of the original separately mis-states the leftover.

60. (C) The reciprocals are 7/3 ≈ 2.33, 8/5 = 1.6, 4/9 ≈ 0.44, and 3/2 = 1.5. The largest reciprocal comes from the smallest fraction, and 3/7 (≈ 0.43) is the smallest of the four, so its reciprocal 7/3 is the largest. Assuming the largest fraction (9/4) gives the largest reciprocal is exactly backwards — the smaller the fraction, the larger its reciprocal.